

Milestone 1 Report

Environment-Division of ISAC for Pinching-Antenna Systems (EDMA-ISAC-PASS)

1.1 Problem Statement

Pinching-antenna systems (PASS) allow a base station to activate radiating elements at flexible positions along a dielectric waveguide, providing near-field, reconfigurable line-of-sight (LoS) links. Sixth-generation (6G) networks additionally require Integrated Sensing and Communication (ISAC), where a single waveform must simultaneously satisfy communication and sensing quality-of-service. Existing PASS-ISAC studies almost universally assume that every candidate antenna position enjoys unobstructed LoS to both the intended user and target. This assumption is unrealistic in practice: obstacles in the propagation environment randomly block LoS links. Also there is no existing study on ISAC PASS blockage model. So my main focus is on building ISAC PASS Blockage model.

1.2 Motivation and Significance

A recent line of work of EDMA in pinching antenna system has shown blockage model for communication-only model, LoS blockage need not be treated purely as an impairment: by placing antennas so that a user's own link stays strong while unwanted links to other users are more likely to be blocked, multiple users can be served simultaneously and interference-free, without precoding or multi-user detection — an approach termed Environment-Division Multiple Access (EDMA). Whether this “blockage-as-a-resource” principle extends to the sensing function, where the analogous impairment is clutter rather than interference, has not previously been examined. Establishing this extension is significant because it would provide a low-complexity, precoding-free physical-layer foundation for dual-functional PASS-ISAC systems.

1.3 Assumptions

- A segmented dielectric waveguide of length D_L , height d , is partitioned into M equal, non-overlapping segments, each with one activated pinching antenna.
- Each segment's antenna is placed at the horizontal midpoint between its dedicated communication user and dedicated sensing target (fixed placement rule).
- LoS existence between any two points is modeled as a Bernoulli random variable with a distance-dependent probability (blockage severity parameter ϕ).
- Interference (communication) and clutter (sensing) contributions are restricted to the two nearest-neighbor segments.
- The antenna's own link is assumed unblocked almost surely, since the midpoint rule confines it to within half a segment width of its own user/target.

1.4 Scope

The current scope covers: (i) closed-form instantaneous and ergodic communication/sensing rate derivation under the nearest-neighbor blockage model, (ii) a formal proof that both ergodic rates are monotonically non-decreasing in ϕ , (iii) closed-form lower bounds on the ergodic sum rates for $M = 2$ and general M , and (iv) Monte Carlo validation against pinching-antenna-assisted and conventional-antenna-assisted TDMA baselines.

2. Review of Existing Work

As identified through this milestone's literature review, no existing PASS-ISAC study explicitly models random LoS blockage as a first-class channel phenomenon affecting both the communication and sensing functions jointly. This is the specific gap our project addresses. The following four works form the core basis for this positioning.

2.1 NTT DOCOMO — “Pinching Antenna” (MWC 2021 technical presentation)

This industry white paper from NTT DOCOMO is the original proposal and hardware demonstration of the pinching-antenna concept. It describes a pinching antenna as a dielectric waveguide (a rod-shaped, coreless, metal-free line similar in construction to optical fiber) that functions as a leaky-wave antenna when small dielectric particles are physically “pinched” onto it, creating one or more radiation points anywhere along the waveguide's length. The paper motivates the technology primarily as a solution for wireless coverage in hard-to-reach or radio-wave-blocked areas, showing that a movable pinch point can dynamically shift the communication area to route around a physical barrier that would otherwise block a fixed base station's signal. It reports a real 60 GHz-band data-transmission test comparing a waveguide with and without an attached pinching antenna, demonstrating that video data is successfully received only when the antenna is pinched onto the waveguide at the receiver's location. This work contains no channel modeling, no stochastic blockage analysis, and no ISAC treatment — it is the foundational hardware proof-of-concept that all subsequent analytical PASS literature (including EDMA and PASS-ISAC works) builds upon.

2.2 “Pinching Antenna Systems for Integrated Sensing and Communications” (IEEE Trans. Wireless Commun., 2026)

This paper proposes a multiple-waveguide PASS-assisted ISAC system in which the base station transmits communication and sensing signals through pinching antennas and collects sensing echoes with a separate uniform linear array (ULA) receiver (the PTUR architecture). It formulates a target-sensing Cramér–Rao Bound (CRB) minimization problem under communication QoS and power-budget constraints, and solves the resulting non-convex problem via an alternating optimization (AO) framework that decomposes it into a digital-beamforming sub-problem (solved with semidefinite relaxation) and a pinching-beamforming sub-problem (solved with a penalty method, successive convex approximation, and element-wise one-dimensional search over PA positions). All wireless channels in this work are modeled as deterministic, always-available LoS channels — no blockage, obstruction, or stochastic geometry of any kind is considered anywhere in the channel model. Simulation results show the proposed framework outperforms massive-MIMO and conventional-MIMO ISAC baselines, and that performance improves with more waveguides and more PAs per waveguide. This is the most comprehensive and recent PASS-ISAC optimization framework identified in our review, and it is also the clearest example of the gap this project addresses: it achieves strong performance precisely because it assumes LoS is always available, an assumption our project explicitly relaxes.

2.3 “Environment Division Multiple Access (EDMA): A Feasibility Study via Pinching Antennas” (IEEE Trans. Wireless Commun., 2026)

This paper introduces EDMA for communication-only, multi-user PASS. Rather than treating random LoS blockage as a pure impairment, it shows that placing each segment's pinching antenna at (or near) its own user's position — so that the desired link stays strong while the links from neighboring segments to that user are more likely to be geometrically and probabilistically blocked — allows multiple users to be served simultaneously and interference-free over a shared frequency band, without any precoding or multi-user detection. The paper derives closed-form outage and ergodic-rate expressions under a Bernoulli LoS-blockage model with nearest-neighbor interference, and proves that performance improves (rather than degrades) as blockage severity

increases, terming this the “blockage-as-a-resource” principle. This is the foundational communication-only result our project directly extends to the joint ISAC setting, where the analogous impairment for the sensing function is clutter rather than interference.

2.4 “LoS Blockage in Pinching-Antenna Systems: Curse or Blessing?” (IEEE Wireless Commun. Lett., 2025)

This letter provides the single-link analytical foundation for the blockage-as-a-resource idea later generalized into EDMA. It analyzes the outage probability of a single pinching-antenna link under random LoS blockage and shows that, depending on the network configuration (in particular, whether blockage primarily removes the desired signal path or an interfering path), increased blockage severity can either degrade or actually improve link reliability — formalizing the dual “curse or blessing” character of blockage in PASS for the first time. It does not consider multi-segment/multi-user interaction or ISAC, both of which are introduced in the EDMA paper and in our project, respectively.

2.5 Identified Gap

Taken together, these four works cover: (i) the physical hardware and coverage motivation for PASS (DOCOMO), (ii) the most advanced PASS-ISAC optimization framework, which assumes deterministic LoS availability throughout (Li et al.), (iii) the blockage-as-a-resource principle for communication-only PASS (EDMA), and (iv) the single-link analytical basis for that principle (Ding and Poor). No existing work — including the most recent and most comprehensive PASS-ISAC paper reviewed above — formulates a PASS-ISAC system under an explicit, stochastic LoS-blockage channel model that jointly governs both the communication and sensing functions. This is the specific, verified gap our project targets.

3. State-of-the-Art (SOTA) Position

We position two works as jointly defining the SOTA boundary this project sits between. First, NTT DOCOMO's original pinching-antenna demonstration established PASS as a physically realizable technology — a movable radiation point on a dielectric waveguide, validated with a real 60 GHz data-transmission test — and is the SOTA for PASS as a hardware architecture; every subsequent analytical work, including our own, adopts this physical model as given. Second, Ding, Schober, and Poor's EDMA feasibility study is the SOTA for exploiting random LoS blockage analytically: it is the strongest existing result showing that blockage can be a provable communication-performance resource rather than only an impairment, via closed-form outage and ergodic-rate analysis under a stochastic blockage model. Our project's SOTA position is defined relative to both: we adopt DOCOMO's physical PASS architecture and extend EDMA's blockage-as-a-resource analytical framework from the communication-only setting to the joint ISAC setting, a combination not present in the Li et al. PASS-ISAC framework (which achieves its results under a deterministic, always-available LoS assumption) or in any other work identified in this review. This SOTA position will be refined as the literature review deepens in subsequent milestones.

4. Project Timeline

Milestone	Planned Work
M1 (current)	Understanding the technology and research of existing work: PASS hardware fundamentals, PASS-ISAC optimization literature, and the EDMA blockage-as-a-resource principle; problem statement and SOTA positioning (this report).
M2	Reproduction: reproduce key existing results from the identified SOTA papers (EDMA ergodic-rate/outage results and, where feasible, the PASS-ISAC CRB baseline) to build a verified implementation foundation.
M3	Problem formulation and deciding system parameters: formally state the joint blockage-aware PASS-ISAC problem and fix the system parameters (carrier frequency, waveguide geometry, blockage severity range, power budget, QoS thresholds) to be used throughout.

M4	Network model and system model: finalize the multi-segment PASS network topology, channel model (blockage + fading), and communication/sensing signal models.
M5	Building the mathematical model: derive closed-form instantaneous and ergodic rate expressions and closed-form bounds under the blockage model established in M4.
M6	Results: numerical evaluation and Monte Carlo validation of the derived closed-form expressions and bounds.
M7	Comparison with baselines and paper writing: benchmark against TDMA and relevant PASS-ISAC baselines, and consolidate all results into the final written report.

References

- [1] NTT DOCOMO, INC., “Pinching Antenna,” MWC 2021 technical presentation.
- [2] H. Li, R. Zhong, Z. Pan, C. Dong, J. Lei, and Y. Liu, “Pinching antenna systems for integrated sensing and communications,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 13416–13429, 2026.
- [3] Z. Ding, R. Schober, and H. V. Poor, “Environment division multiple access (EDMA): A feasibility study via pinching antennas,” *IEEE Trans. Wireless Commun.*, vol. 25, pp. 15675–15691, 2026.
- [4] Z. Ding and H. V. Poor, “LoS blockage in pinching-antenna systems: Curse or blessing?,” *IEEE Wireless Commun. Lett.*, vol. 14, no. 9, pp. 2798–2802, 2025.